

Friction

Friction force is of two types: (a) Kinetic, (b) Static.

Kinetic Friction

$$f_k = \mu_k \times N$$

The proportionality constant μ_k is called the coefficient of kinetic friction and its value depends on the nature of the two surfaces in contact.

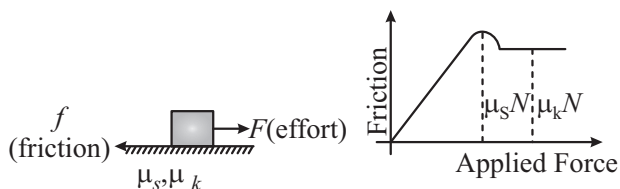
Static Friction

It exists between the two surfaces when there is tendency of relative motion but no relative motion along the two contact surfaces.

This means static friction is a variable and self adjusting force. However it has a maximum value called limiting friction.

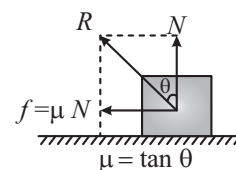
$$f_{\max} = \mu_s N$$

$$0 < f_s < f_{\max}$$



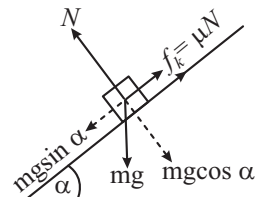
Angle of Friction

It is the angle which the resultant of the force of limiting friction and the normal reaction (N) makes with the direction of N .



Angle of Repose or Angle of Sliding

It is the angle of inclination of a plane with the horizontal, such that a body placed on it, just begins to slide down.



If angle of repose is α and coefficient of limiting friction is μ_s , then

$$\mu_s = \tan \alpha$$